USN | | | | | | | | | | |

[24BSACH03]

FIRST SEMESTER B. Tech EXAMINATION, JANUARY- 2024

APPLIED CHEMISTRY

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

			Blooms Level	CO	Marks
		PART-A	**5 X 2=10**		
1.	**a.**	Define Standard reduction potential (E^o)?	L-1	CO1	**02**
	b.	Explain internal energy.	L-1	CO2	**02**
	c.	What is effective nuclear charge?	L-2	CO3	**02**
	d.	Describe optical activity.	L-3	CO4	**02**
	e.	What are nanomaterials?	L-1	CO5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Discuss the construction and working principle of standard hydrogen electrode.	L-2	CO1	**05**
		OR			
	b.	Differentiate between galvanic cell and electrolytic cell.	L-2	CO1	**05**
3.	**a.**	Articulate how different variables impact on the equilibrium constant.	L-3	CO2	**05**
		OR			
	b.	List out the difference between reaction quotient and equilibrium constant.	L-2	CO2	**05**
4.	**a.**	Illustrate the atomic size and electronegativity along the periods and groups.	L-3	CO3	**05**
		OR			
	b.	Draw the molecular orbital diagram of O_2 and calculate the bond order.	L-3	CO3	**05**
5.	a.	What is structural and geometrical isomerism? Give example.	L-2	CO4	**05**
		OR			
	b.	Define substitution reaction and explain SN_2 reaction mechanism.	L-1	CO4	**05**
6.	**a.**	Describe the working principle of XRD instrument	L-3	CO5	**05**
		OR			
	b.	Summarize the futuristic applications of nanomaterials.	L-2	CO5	**05**

Deemed to be University
Learning Resource Centre
Jain Global Campus
Ramanagara-112

7.	a.	Summarize the differential metal and differential aeration corrosion.	L-2	CO5	05
		OR			
	b.	Extract the bond order and magnetic behavior of N_2 molecule using molecular orbital diagram.	L-2	CO5	05
		PART-C	**6 X 10=60**		
8.	a.	Derive Nernst's equation for the single electrode reaction.	L-2	CO1	10
		OR			
	b.	Discuss the factors affecting on rate of corrosion.	L-2	CO1	10
9.	a.	Illustrate the characteristics of Ellingham diagram with neat labeled diagram.	L-1	CO2	10
		OR			
	b.	Summarize the contact procedure for synthesizing H_2SO_4 and explain Le-principle Chatelier's in its synthesis.	L-3	CO2	10
10.	a.	Explain briefly about classification of elements in modern periodic table.	L-2	CO3	10
		OR			
	b.	Extract the bond order, magnetic behavior and molecular orbital configuration of CO using molecular orbital diagram.	L-3	CO3	10
11.	a.	What is conformational analysis? Discuss the conformational analysis of ethane.	L-3	CO4	10
		OR			
	b.	How do you prepare Paracetamol and Acetylsalicylic acid (Aspirin).Write the reaction?	L-3	CO4	10
12.	a.	Explain the synthesis of nanoparticles by using sol- gel method.	L-2	CO5	10
		OR			
	b.	Summarize the principle, working mechanism and applications of Fourier transform infrared spectroscopy.	L-3	CO5	10
13.	a.	What is crystal field splitting? Explain octahedral splitting in complexes.	L2	CO3	10
		OR			
	b.	Define the term conformational analysis and discuss the conformational analysis of cyclohexane molecule.	L-1	CO1	10